

DBMS Curriculum

Introduction to RDBMS

Understanding Databases

- **Introduction to Databases**
 - Definition and history
 - Importance of data management
- **Types of Databases**
 - Relational vs. Non-relational databases
 - Overview of SQL and NoSQL

Fundamentals of RDBMS

- **Key Concepts**
 - Tables, rows, and columns
 - Primary keys and foreign keys
 - Indexes
- **Introduction to SQL**
 - Basic syntax
 - Data types

Working with SQL

Data Manipulation Language (DML)

- **CRUD Operations**
 - SELECT, INSERT, UPDATE, DELETE
- **Filtering and Sorting Data**

- WHERE clause, ORDER BY

Data Definition Language (DDL)

- **Creating and Modifying Schema**

- CREATE, ALTER, DROP

Advanced SQL Concepts

- **Joins and Relationships**

- INNER JOIN, LEFT JOIN, RIGHT JOIN, FULL JOIN

- **Subqueries and Nested Queries**

- **Aggregation and Grouping**

- GROUP BY, HAVING

Database Design and Normalisation

Database Design Principles

- **Understanding Normalisation**

- Purpose and goals
- Normal forms (1NF, 2NF, 3NF, BCNF)

- **ER Diagrams and Schema Design**

- Designing a database schema
- Entity-Relationship (ER) modeling

Advanced RDBMS Topics

Transaction Management

- **ACID Properties**

- **Concurrency Control**

- **Locks and Deadlocks**

Indexing and Optimisation

- **Index Types and Usage**

- **Query Optimisation Techniques**
- **Performance Tuning**

Practical Applications and Projects

Implementing a Database

- **Setting up a Database Server**
 - Installation and configuration (e.g., MySQL, PostgreSQL)
- **Database Security**
 - Authentication and authorisation

Real-world Projects

- **Project Ideas**
 - E-commerce database design
 - Inventory management system
 - Student information system
- **Implementation Steps**
 - Requirements analysis
 - Schema design and implementation
 - Data insertion and queries